

Publications

Journal Papers

- Lavagna, G. Buttitta, S. Bonacorsi, C. Barbarito, M. Moliterno, G. Saito, I. Oddone, G. Verdone, S. Raimondi & Panella, M.**
Smart Microfluidics: a curated dataset of microfluidic liposome formulations with cross-laboratory validation for machine-learning applications
Data in Brief, **112667**, 2026.
<https://doi.org/10.1016/j.dib.2026.112667>
 Keywords: Drug delivery, Nanoparticles, Dynamic light scattering, Machine learning.
- Lavagna, L., Carillo, S., & Panella, M.**
A topical review on time-independent perturbation theory in one-dimensional quantum systems.
Physica Scripta, **100** (10), 1–33, 2025.
<https://doi.org/10.1088/1402-4896/ae0a8f>
 Keywords: one-dimensional quantum systems; perturbation theory; quantum computing.
- Lavagna, L., Piperno, S., Ceschini, A., & Panella, M.**
Small graph perturbations, QAOA, and the MaxCut problem.
AVS Quantum Science, 1–14, 2025.
 Publisher: AIP Publishing; AVS Science and Technology of Materials, Interfaces and Processing.
 ISSN: 2639-0213.
 Keywords: quantum approximate optimization algorithm; MaxCut problem; graph perturbations; quantum computing.
- Buttitta, G., Lavagna, L., Bonacorsi, S., Barbarito, C., Moliterno, M., Saito, G., Oddone, I., Verdone, G., Raimondi, S., & Panella, M.**
Machine Learning-Guided microfluidic optimization of clinically inspired liposomes for nanomedicine applications.
International Journal of Pharmaceutics, **686**, 126362, 2025.
<https://doi.org/10.1016/j.ijpharm.2025.126362>
 Keywords: machine learning; artificial intelligence; liposomes; microfluidics; nanomaterial; nanomedicine.

Conference Papers

- L. Lavagna, G. Vittori, A. Rosato & M. Panella**
Additive Resource Scaling in Quantum Circuits for Search in Cryptanalysis
 In *Proceedings of the IEEE International Symposium on Circuits and Systems (ISCAS 2026)*, IEEE, pp. 2758–2762, 2026.
<https://doi.org/10.1109/ISCAS66217.2026.11562218> Keywords: circuits and systems; quantum cryptography.
- L. Lavagna, G. Vittori, A. Rosato & M. Panella**
A Unified Quantum Circuit Framework for Grover Search and Quantum Walks in NISQ Cryptanalysis
 In *Proceedings of the IEEE International Symposium on Circuits and Systems (ISCAS 2026)*, IEEE,

pp. 837-841, 2026.

<https://doi.org/10.1109/ISCAS66217.2026.11562366>

Keywords: circuits and systems; quantum cryptography.

- **M. Casalbore, L. Lavagna, A. Rosato and M. Panella**

Time Series Anomaly Detection with Quantum Variational Methods and Set Covering

In *Proceedings of the 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. Barcelona, Spain, 2026, pp. 1846-1850

doi: 10.1109/ICASSP55912.2026.11464824

Keywords: quantum approximate optimization algorithm; quantum computing; time series.

- **F. De Falco, L. Lavagna, A. Ceschini, & M. Panella**

Enhancing QAOA Ansatz via Multi-Parameterized Layer and Blockwise Optimization

In *Proceedings of the 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2026)*. Barcelona, Spain, 2026, pp. 20417-20421,

<https://doi.org/10.1109/ICASSP55912.2026.11461345>

Keywords: quantum approximate optimization algorithm; quantum computing.

- **Lavagna, L., Ceschini, A., Rosato, A., & Panella, M.**

Novel Quantum Approaches to Hyperdimensional Computing for Neural Networks.

In *Proceedings of the International Joint Conference on Neural Networks (IJCNN 2025)*, Rome, Italy, IEEE, pp. 1-8, 2025.

<https://doi.org/10.1109/IJCNN64981.2025.11229083>

Keywords: quantum computing; accuracy; computational modeling; neural networks; machine learning; computer architecture; computational efficiency; quantum circuit; testing.

- **Casalbore, M., Lavagna, L., Rosato, A., & Panella, M.**

Hybrid Quantum-Classical Framework for Anomaly Detection in Time Series with QUBO formulation and QAOA.

In *Proceedings of the International Joint Conference on Neural Networks (IJCNN 2025)*, Rome, Italy, IEEE, pp. 1-8, 2025.

<https://doi.org/10.1109/IJCNN64981.2025.11228152>

Keywords: statistical analysis; time series analysis; pipelines; qubit; noise; computer architecture; quantum circuit; anomaly detection; optimization; tuning.

- **Lavagna, L., De Falco, F., Ceschini, A., Rosato, A., & Panella, M.**

Trade-offs in Cryptosystems by Boolean and Quantum Circuits.

In *Proceedings of the IEEE International Symposium on Circuits and Systems (ISCAS 2025)*, IEEE, pp. 1-5, 2025.

<https://doi.org/10.1109/ISCAS56072.2025.11043205>

Keywords: fault tolerance; circuits and systems; encryption; quantum mechanics; circuit theory.

- **De Falco, F., Lavagna, L., Ceschini, A., Rosato, A., & Panella, M.**

Evolving hybrid quantum-classical GRU architectures for multivariate time series.

In *IEEE International Workshop on Machine Learning for Signal Processing (MLSP 2024)*, IEEE Computer Society, pp. 1-6, 2024.

<https://doi.org/10.1109/MLSP58920.2024.10734792>

Keywords: multivariate time series; quantum computing; quantum gated recurrent units; quantum machine learning.

- **Lavagna, L., Ceschini, A., Rosato, A., & Panella, M.**

A layerwise-multi-angle approach to fine-tuning the quantum approximate optimization algorithm.
In *Proceedings of the International Joint Conference on Neural Networks (IJCNN 2024)*, IEEE, pp. 1–6, 2024.

<https://doi.org/10.1109/IJCNN60899.2024.10650075>

Keywords: quantum approximate optimization algorithm; layerwise-multi-angle approach; quantum computing.

Other

- **Cacioppo, A., De Falco, F., Di Luzio, F., Giagu, S., Lavagna, L. & Panella, M.**

Quantum-Enhanced Fraud Detection: A Comparative Study on Real-World Financial Data In *3rd Workshop on Quantum Computing @ INFN Milano, Italia (02/2026)*.

- **Lavagna, L., De Falco, F., & Panella, M.**

Quantum Hyperdimensional Computing for Pattern Completion.

In *Proceedings of Quantum Techniques in Machine Learning (QTM 2025)*, pp. 1–1, 2025.

- **Lavagna, L., De Falco, F., & Panella, M.**

The Effectiveness of Classical and Hybrid Models for MaxCut problem.

In *Proceedings of Quantum Techniques in Machine Learning (QTM 2025)*, pp. 1–1, 2025.

- **Lavagna, L., De Falco, F., & Panella, M.**

Is the QAOA the Ultimate Solution for the MaxCut problem?

In *Proceedings of Quantum Techniques in Machine Learning (QTM 2025)*, pp. 1–1, 2025.

- **Ceschini, A., Lavagna, L., De Falco, F., Rosato, A., & Panella, M.**

Circuiti neurali quantistici per il processing di grafi, immagini e serie temporali.

In *Memorie ET2025*, Gruppo Nazionale Ricercatori di Elettrotecnica, Italia, pp. 1–2, 2025.

- **Lavagna, L., Ceschini, A., Piperno, S., Casalbore, M., Rosato, A., & Panella, M.**

Soluzioni quantistico-classiche per ottimizzazione e rilevamento di anomalie.

In *Memorie ET2025*, Gruppo Nazionale Ricercatori di Elettrotecnica, Italia, pp. 1–2, 2025.

- **Rosato, A., Lavagna, L., & Panella, M.**

Integrazione del calcolo iperdimensionale nei circuiti digitali, nelle reti neurali e nelle architetture computazionali quantistiche.

In *Memorie ET2025*, Gruppo Nazionale Ricercatori di Elettrotecnica, Italia, pp. 1–2, 2025.

- **Ceschini, A., Lavagna, L., De Falco, F., Rosato, A., & Panella, M.**

Convergenza e generalizzazione nelle reti neurali quantistiche.

In *Memorie ET2024*, Gruppo Nazionale Ricercatori di Elettrotecnica, Italia, pp. 1–2, 2024.

- **De Falco, F., Piperno, S., Lavagna, L., Ceschini, A., Rosato, A., & Panella, M.**

Enhancing QAOA Ansatz via Multi-Parameterized Layer and Blockwise Optimization.

In *Proceedings of Quantum Techniques in Machine Learning (QTM 2024)*, pp. 1–3, 2024.

- **Lavagna, L., De Falco, F., Piperno, S., Ceschini, A., Rosato, A., & Panella, M.**

Quantum Generative Modeling via Straightforward State Preparation.

In *Proceedings of Quantum Techniques in Machine Learning (QTML 2024)*, University of Melbourne, Melbourne, Australia, pp. 1–1, 2024.

- **Piperno, S., Lavagna, L., De Falco, F., Ceschini, A., Rosato, A., Windridge, D., & Panella, M.**
Quantum Enhanced Knowledge Distillation.
In *Proceedings of Quantum Techniques in Machine Learning (QTML 2024)*, University of Melbourne, Melbourne, Australia, pp. 1–1, 2024.